Optimization-based ILC design

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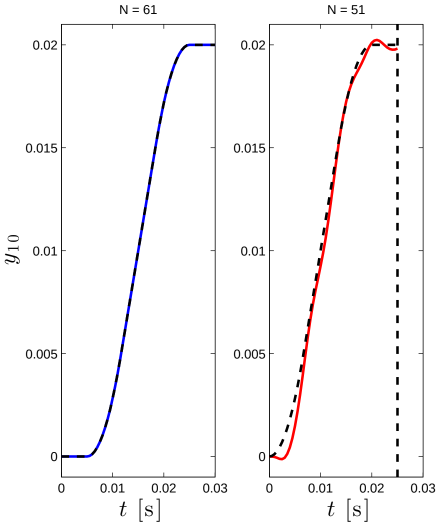
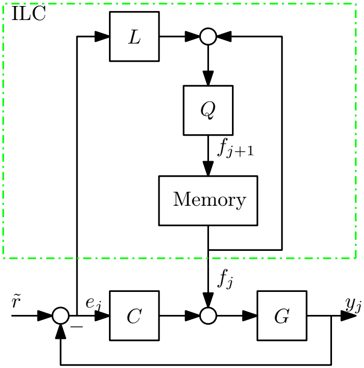
Example: frequency domain ILC



Result

- $L = (GS)^{-1}$, $Q = 1$
- faster trajectory
- 10 iterations
- trial length N
  - $N = 61$: expected result
  - $N = 51$: what's wrong?

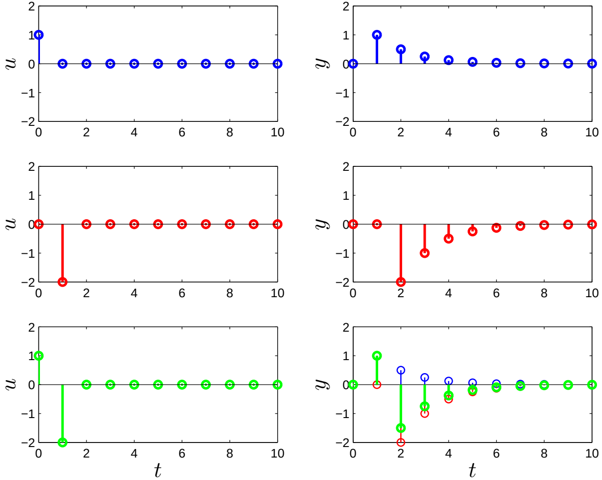
ILC scheme



ILC summary

- learning update: $f_{j+1} = Q(f_j + Le_j)$
- frequency domain design procedure
  - $L = (GS)^{-1}$
  - $|Q(1 - GSL)| < 1, \quad \forall \omega \in [0, 2\pi]$
- underlying assumptions
  - infinite time
  - LTI
  - SISO (mostly)
- this part:
  - finite time
  - LTI & LTV
  - optimal
  - mainly MIMO
  - basis functions
  - ...

Intuitive idea



$$\begin{bmatrix} y(0) \\ y(1) \\ \vdots \\ y(10) \end{bmatrix} = \begin{bmatrix} h_0 \\ h_1 \\ \vdots \\ h_{10} \end{bmatrix} u(0)$$

$$\begin{bmatrix} y(0) \\ y(1) \\ \vdots \\ y(10) \end{bmatrix} = \begin{bmatrix} 0 \\ h_0 \\ \vdots \\ h_9 \end{bmatrix} u(1)$$

$$\begin{bmatrix} y(0) \\ y(1) \\ \vdots \\ y(10) \end{bmatrix} = \begin{bmatrix} h_0 & 0 \\ h_1 & h_0 \\ \vdots & \\ h_{10} & h_9 \end{bmatrix} \begin{bmatrix} u(0) \\ u(1) \end{bmatrix}$$

Convolution causal system

$$y(k) = \sum_{l=0}^{\infty} h_l u(k-l) \quad u(k) = 0, k < 0$$

- $y(0) = h_0 u(0)$
- $y(1) = h_0 u(1) + h_1 u(0)$
- $y(2) = h_0 u(2) + h_1 u(1) + h_2 u(0)$
- $y(3) = h_0 u(3) + h_1 u(2) + h_2 u(1) + h_3 u(0)$

Convolution matrix

$$\begin{bmatrix} y(0) \\ y(1) \\ y(2) \\ y(3) \end{bmatrix} = \underbrace{\begin{bmatrix} h_0 & 0 & 0 & 0 \\ h_1 & h_0 & 0 & 0 \\ h_2 & h_1 & h_0 & 0 \\ h_3 & h_2 & h_1 & h_0 \end{bmatrix}}_{\text{convolution matrix}} \begin{bmatrix} u(0) \\ u(1) \\ u(2) \\ u(3) \end{bmatrix}$$

Basic Finite Time ILC Framework

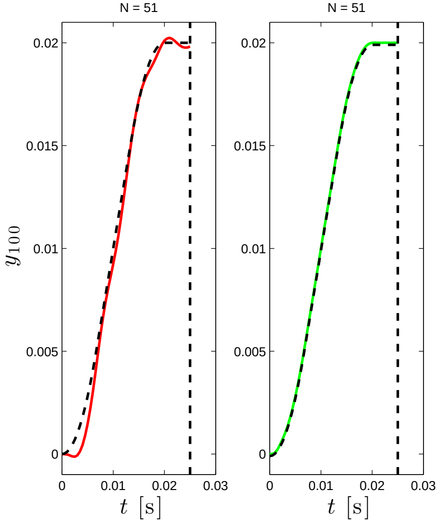
Frequency domain ILC controller

- $f_{j+1} = Q(f_j + Le_j)$
- set $L = (GS)^{-1}$
- set $Q = 1$

Intuitive lifted ILC controller

- $f_{j+1} = f_j + Le_j$
- Tikhonov regularized inverse
  - $L = (J^T J + \rho I)^{-1} J^T$
  - if J regular and $\rho = 0$, then $L = J^{-1}$
  - also known as ridge regression in statistics

Example revisited: frequency domain ILC

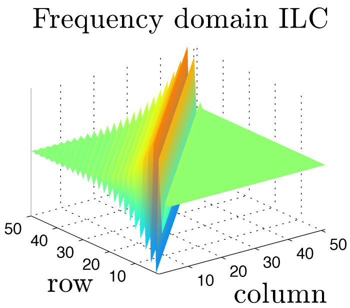


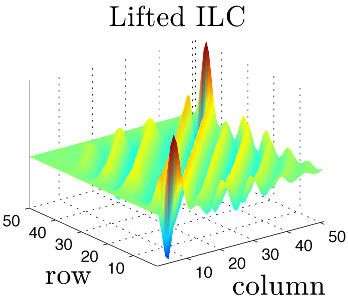
Result after many iterations

- Frequency domain: large error
- lifted ILC: small error...!

Example revisited: frequency domain ILC

Toeplitz matrix of L:

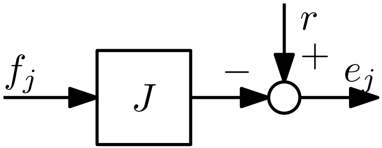




- Important difference: edge effects in red and blue
- Note: frequency domain ILC uses ZPETC

System relations

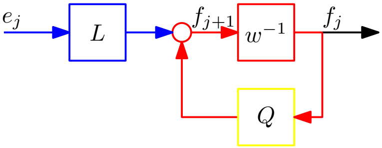
- system
  - $e_j = r - Jf_j$
  - J square, lower triangular and Toeplitz
    - time invariant
    - causal



  - note: in earlier frequency domain approach $e_j = S\tilde{r} - GSf_j$
    $\Rightarrow$ take $r = S\tilde{r}$, $J = GS$

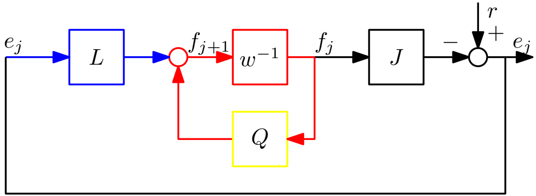
System relations

- ILC controller
  - $f_{j+1} = Qf_j + Le_j$
  - note: different definition Q compared to frequency domain ILC
    - more convenient for synthesis (Slide 21)
    - no fundamental difference - also both are used in literature
  - Q, L not necessarily lower triangular and Toeplitz
    - time varying
    - non-causality



System relations

- system $e_j = r - Jf_j$
- ILC controller $f_{j+1} = Qf_j + Le_j$



- system relations resemble state space

$$
\begin{aligned}
f_{j+1} &= (Q - LJ)\, f_j + Lr \\
e_j &= \qquad\quad -Jf_j + r
\end{aligned}
$$

- system is asymptotically stable if $|\lambda_i(Q - LJ)| < 1, \quad i = 1, 2, \ldots, Nl$

ILC analysis

- recall state-space system

$$\begin{aligned} f_{j+1} &= (Q - LJ)\, f_j + Lr \\ e_j &= \qquad -Jf_j + r \end{aligned}$$

- state-space to iteration-domain transfer function matrix $r \mapsto e_j$

$$e_j = \left(I_{Nl} - J(wI_{Nl} - Q + LJ)^{-1} L \right) r$$

- steady-state error: set $w = 1$

$$e_\infty = \left(I_{Nl} - J(I_{Nl} - Q + LJ)^{-1} L \right) r$$

- let $Q = I_{Nl}$.
  - then $e_\infty = \left(I_{Nl} - J(LJ)^{-1} L \right) r$
  - $e_\infty = 0$ requires LJ to be non-singular $\Rightarrow$ J and L square and non-singular

Singularity of J

- reasons:
  - strictly proper systems or additional delays: if $h_0, h_1, \ldots h_{d-1} = 0$, then $\mathrm{rank}(J)$ is at most $Nm - d$
  - non-minimum phase zeros
- consequences:
  - no full stabilization
  - no full rejection

Example

$$h = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \quad \Rightarrow \quad J = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ -1 & 1 & 0 \end{bmatrix}, \quad r = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$$

Thus: $e_\infty(1) = 1$

Convergence properties revisited

- $e_j = r - Jf_j$
- $f_{j+1} = Qf_j + Le_j = (Q - LJ)f_j + Lr$
- asymptotically stable if $|\lambda_i(Q - LJ)| < 1, \quad i = 1, 2, \ldots, Nl$

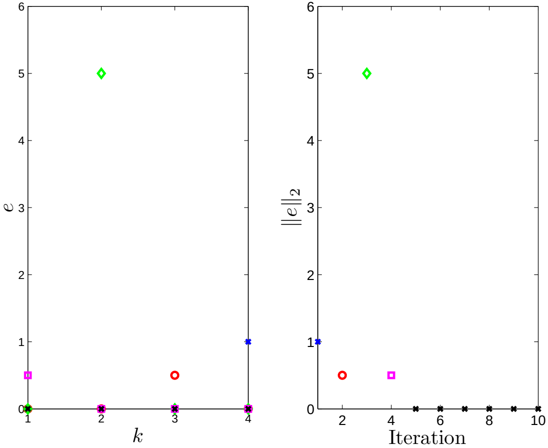
Example

$$Q = I_4, L = \begin{bmatrix} 1 & -\frac{1}{10} & 0 & 0 \\ 0 & 1 & -10 & 0 \\ 0 & 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix}, J = I_4$$

- Q: asymptotically stable?
  A: $\lambda_i(Q - LJ) = 0, \; i = 1, 2, 3, 4 \Rightarrow$ asymptotically stable!

Example

- $f_1 = 0$
- $e_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$
- error increases at iteration 3
- but indeed converges to $e_\infty = 0$
- undesired transient behavior



ILC learning transient

- no guarantees provided by convergence condition $\lambda_i(Q - LJ) = 0, \; i = 1, 2, 3, 4$
- of essential importance in practice

Result

Monotonic convergence of f_j in the ℓ_p norm, i.e.,

$$\|f_\infty - f_{j+1}\|_p \leq k\|f_\infty - f_j\|_p$$

for some $k \in [0, 1)$, is achieved if

$$\|Q - LJ\|_{p,p} < 1$$

Note: can also be defined for e_j

Convergence conditions

- convergence: $\max_i |\lambda_i(Q - LJ)| < 1$
- monotonic convergence: $\|Q - LJ\|_{p,p} < 1$

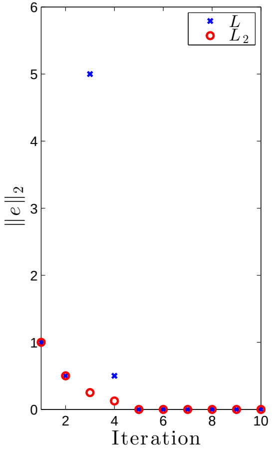
Convergence results

- spectral radius $\max_i |\lambda_i(Q - LJ)|$ is not a norm
- for any (matrix) norm $\|.\|$: $\max_i |\lambda_i(Q - LJ)| \leq \|Q - LJ\|$
- hence: monotonic convergence $\Rightarrow$ convergence
- monotonic convergence in 2-norm obtained if $\bar{\sigma}(Q - LJ) < 1$
- convergence in previous part (frequency domain): also monotonic

Example revisited

- $L = \begin{bmatrix} 1 & -\frac{1}{10} & 0 & 0 \\ 0 & 1 & -10 & 0 \\ 0 & 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix}$
- $\max_i |\lambda_i(Q - LJ)| = 0 \Rightarrow$ convergence
- $\bar{\sigma}(Q - LJ) = 10 \Rightarrow$ no monotonic convergence

- $L_2 = \begin{bmatrix} 1 & -\frac{1}{2} & 0 & 0 \\ 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix}$
- $\max_i |\lambda_i(Q - L_2J)| = 0 \Rightarrow$ convergence
- $\bar{\sigma}(Q - L_2J) = \frac{1}{2} \Rightarrow$ monotonic convergence



Optimal ILC synthesis

- system relations

$$
\begin{aligned}
f_{j+1} &= Qf_j + Le_j \\
e_j &= r - Jf_j
\end{aligned}
$$

- how to design Q and L?
- formalization of Tikhonov regularization on slide 6
- consider criterion

$$\mathcal{J}(f_{j+1}) = e_{j+1}^T W_e e_{j+1} + f_{j+1}^T W_f f_{j+1} + (f_{j+1} - f_j)^T W_{\Delta f}(f_{j+1} - f_j)$$

- goal: compute $\arg\min_{f_{j+1}} \mathcal{J}(f_{j+1})$
- weights W_e, W_f, $W_{\Delta f}$:
  - symmetric, positive (semi) definite
  - typical choice: $W_e = I$, $W_{\Delta f} = w_{\Delta f} I$, $W_f = w_f I$, with $w_f, w_{\Delta f} \geq 0$

Reformulate

$$e_j = r - Jf_j \quad \Rightarrow \quad r = e_j + Jf_j$$

Hence

$$\begin{aligned} e_{j+1} &= r - Jf_{j+1} \\ &= e_j - J(f_{j+1} - f_j) \end{aligned}$$

and

$$\begin{aligned} \mathcal{J}(f_{j+1}) =& (e_j - J(f_{j+1} - f_j))^T W_e (e_j - J(f_{j+1} - f_j)) \\ &+ f_{j+1}^T W_f f_{j+1} + (f_{j+1} - f_j)^T W_{\Delta f} (f_{j+1} - f_j) \end{aligned}$$

Optimal ILC synthesis

Optimal ILC

- first order necessary condition for optimality: $\frac{\partial \mathcal{J}(f_{j+1})}{\partial f_{j+1}} = 0$

Result

- use identity $\frac{\partial (s-Ax)^T W(s-Ax)}{\partial x} = -2A^T W(s - Ax)$
- then $\frac{\partial \mathcal{J}(f_{j+1})}{\partial f_{j+1}} = 0 \Rightarrow$ implies

$$(J^T W_e J + W_f + W_{\Delta f})f_{j+1} - (J^T W_e J + W_{\Delta f})f_j - J^T W_e e_j = 0$$

$$f_{j+1} = \underbrace{(J^T W_e J + W_f + W_{\Delta f})^{-1}(J^T W_e J + W_{\Delta f})}_{Q} f_j + \underbrace{(J^T W_e J + W_f + W_{\Delta f})^{-1} J^T W_e}_{L} e_j$$

Analysis of optimal ILC

- monotonic convergence:
  $\bar{\sigma}(Q - LJ) = \bar{\sigma}\left((J^T W_e J + W_f + W_{\Delta f})^{-1} W_{\Delta f}\right) < 1$
- result: since $J^T W_e J + W_f \succeq 0$

1. if $J^T W_e J + W_f \succ 0$, i.e., full rank, then guaranteed monotonic convergence
2. if J singular, then set $W_f \succ 0$ to guarantee monotonic convergence
3. if $W_{\Delta f} = 0$, then deadbeat ILC
4. if J non-singular, then $W_f = W_{\Delta f} = 0$ and $W_e = w_e I$ leads to

   $f_{j+1} = f_j + (J^T J)^{-1} J^T e_j = f_j + J^{-1} e_j \Rightarrow$ inverse model ILC

Optimal ILC synthesis: Analysis

Steady state error

$$e_\infty = \left(I_{Nl} - J(I_{Nl} - Q + LJ)^{-1} L \right) r = \left(I_{Nl} - J(J^T W_e J + W_f)^{-1} J^T W_e \right) r$$

Design guidelines

- if J non-singular, then $W_f = 0$ leads to $e_\infty = 0$
- can be shown that $W_f \succ 0$ leads to improved robustness w.r.t. model uncertainty
- $W_{\Delta f}$:
  - coincides with specific Tikhonov regularization on slide 6
  - large: attenuate trial-variant disturbances
  - small: faster convergence in trial-invariant disturbance situation

Basic Finite Time ILC Framework

ILC analysis

Optimal ILC synthesis

Design Procedure

Low-Order ILC

Summary

Design Procedure

1. Obtain an impulse response of system
   - from a given parametric model
   - from experiments
     - impulse response testing
     - estimation of FIR model using general inputs (fairly easy)

2. Compute Toeplitz matrix J

3. Select tuning parameters W_e, W_f, and $W_{\Delta f}$

4. Compute optimal ILC: L and Q on slide 21

Basic Finite Time ILC Framework

ILC analysis

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Design Procedure

Low-Order ILC

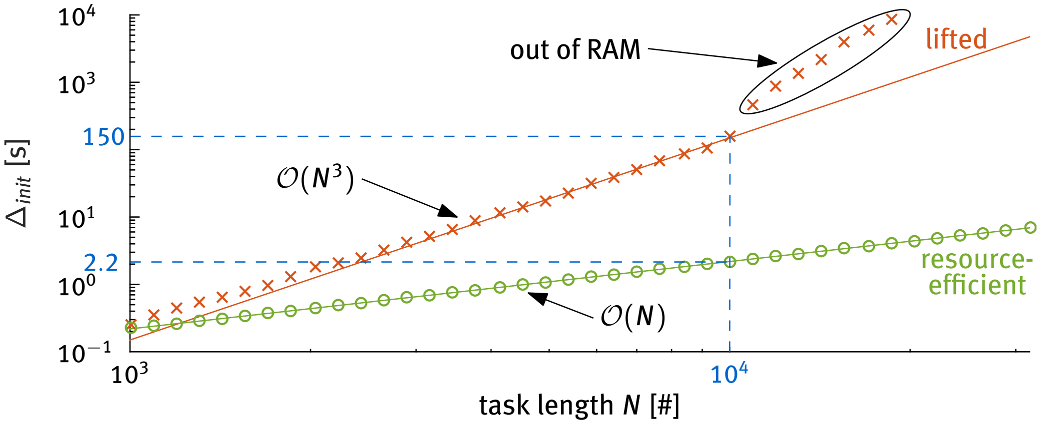
Summary

Low-order implementation of optimal ILC controller of slide 21 (Oomen et al.2011) (van Zundert et al.2016)

$$f_{j+1} = C^{\text{ILC}}(k) \begin{bmatrix} e_j(t) \\ f_j(k) \\ g_j(k+1) \end{bmatrix}$$

- state-space $C^{\text{ILC}}(k)$ follows from Riccati difference equation
- related to LQ tracking problem: $g_{<k>}(k+1)$ noncausal
  - generalization of (Amann et al.1996)
- tractable for large scale problems
  - matrix dimensions invariant under
    - intersample behavior
    - $\#n_u, \#n_y$: MIMO
    - N: trial length
  - in contrast: approach on slide 21 inflates matrix dimensions

Low-order implementation of slide 21 (Oomen et al.2011) (van Zundert et al.2016)



Key point: computationally faster ILC implementation with identical control performance

Basic Finite Time ILC Framework

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Low-Order ILC

Summary

Summary

Lifted ILC

- finite time, LTI/LTV, MIMO
- general framework for repetitive control and ILC (de Roover et al.2000)

Analysis using linear algebra

- convergence vs. monotonic convergence

Synthesis

- optimal ILC algorithms

Extensions

- many different extensions and applications
  - see overview papers, e.g., (Bristow et al.2006), (Blanken et al.2016)
  - many contributions of CST M.Sc./Ph.D. students, e.g., `http://www.toomen.eu`
  - `https://youtu.be/kj_ouy1Fnko?feature=shared` for task-flexible ILC

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